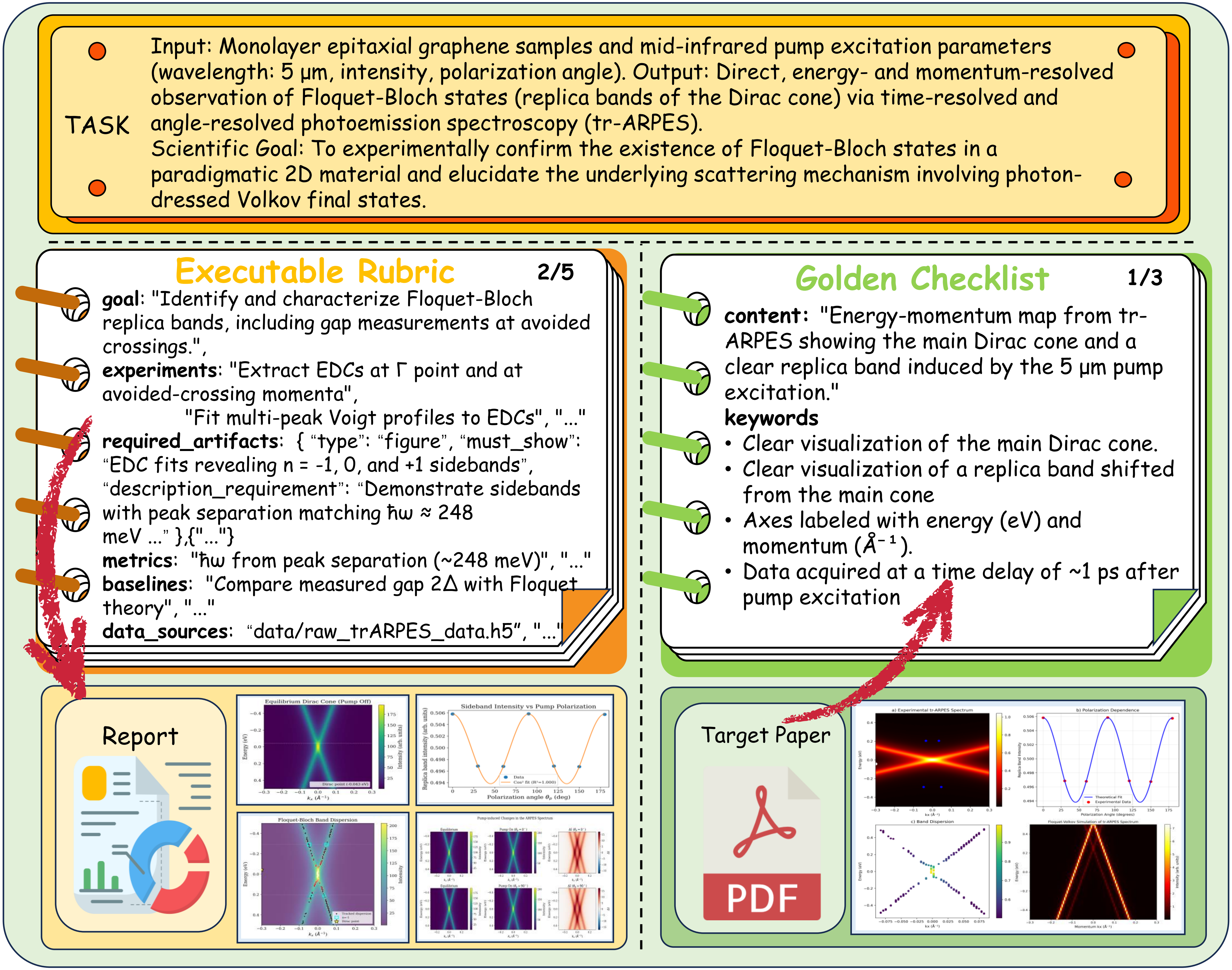
TASK
Input: Monolayer epitaxial graphene samples and mid-infrared pump excitation parameters (wavelength: 5 μm, intensity, polarization angle). Output: Direct, energy- and momentum-resolved observation of Floquet-Bloch states (replica bands of the Dirac cone) via time-resolved and angle-resolved photoemission spectroscopy (tr-ARPES).
Scientific Goal: To experimentally confirm the existence of Floquet-Bloch states in a paradigmatic 2D material and elucidate the underlying scattering mechanism involving photon-dressed Volkov final states.
Executable Rubric
2/5
goal: "Identify and characterize Floquet-Bloch replica bands, including gap measurements at avoided crossings.",
experiments: "Extract EDCs at Γ point and at avoided-crossing momenta",
"Fit multi-peak Voigt profiles to EDCs", "..."
required_artifacts: { "type": "figure", "must_show": "EDC fits revealing n = -1, 0, and +1 sidebands", "description_requirement": "Demonstrate sidebands with peak separation matching ℏω ≈ 248 meV ..." },{"..."}
metrics: "ℏω from peak separation (~248 meV)", "..."
baselines: "Compare measured gap 2Δ with Floquet theory", "..."
data_sources: "data/raw_trARPES_data.h5", "..."
Golden Checklist
1/3
content: "Energy-momentum map from tr-ARPES showing the main Dirac cone and a clear replica band induced by the 5 μm pump excitation."
keywords
• Clear visualization of the main Dirac cone.
• Clear visualization of a replica band shifted from the main cone
• Axes labeled with energy (eV) and momentum (Å⁻¹).
• Data acquired at a time delay of ~1 ps after pump excitation
Report
Equilibrium Dirac Cone (Pump Off)
Sideband Intensity vs Pump Polarization
Floquet-Bloch Band Dispersion
Pump-induced Changes in the ARPES Spectrum
Target Paper
PDF
a) Experimental tr-ARPES Spectrum
b) Polarization Dependence
c) Band Dispersion
Floquet-Volkov Simulation of tr-ARPES Spectrum